Panel Data Analysis

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What is panel data?

- **Cross-section:** many units i, each observed **once**.
- **Time series: one** unit, many periods t (e.g. French GDP over time).
- **Panel (longitudinal):** the **same** units i followed over periods t — at least **two** dimensions.

Higher dimensions are common:

- exports between i and j, at time t, for product k;
- wages of workers i in sectors k, cities c, over years t.

The problem: what the error hides

We want the causal effect β of x_{it} on y_{it}. Run a naive regression and everything we cannot measure lands in the error:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} = \underbrace{f_i}_{\text{individual}} + \underbrace{f_t}_{\text{time}} + u_{it}.$$

- f_i — **unobserved individual** characteristics: ability, institutions, geography… fixed over time, differing across units.
- f_t — **time** effect: shocks hitting **all** units in period t (crises, world prices, technology).
- If f_i or f_t correlates with x_{it}, OLS is **biased** (omitted-variable bias).

The promise of panel data

Observing the *same* units over *several* periods lets us **sweep f_i and f_t out of the error** — identifying β even when they correlate with x.

Balanced vs. unbalanced panels

- **Balanced:** every unit i is observed in **every** period t — exactly $N \times T$ rows (a full rectangle).
- **Unbalanced:** some unit–period cells are **missing** (units enter/exit, attrition) — fewer than $N \times T$ rows.

Why it matters

A couple of results below — notably that the **between** estimator removes the time effect — hold cleanly **only for balanced panels**. Real data are often unbalanced; we flag where it bites.

Roadmap

1. Panel data: dimensions and the model.
2. Estimators that **ignore** heterogeneity: pooled OLS and between.
3. The **within (fixed-effects)** estimator.
4. **Two-way fixed effects**.
5. Application: temperature and growth (Dell et al. 2012).

We deliberately skip random effects and the Hausman test. Random effects requires $\mathrm{Cov}(x_{it}, f_i) = 0$ — the very assumption panel data let us **avoid** — so it is rarely defensible in modern applied work.

Section 1

Naive estimators

OLS in matrix form: one engine, four datasets

Stack all NT observations. With outcome vector y ($NT \times 1$) and regressor matrix X ($NT \times k$), every model in this lecture is the **same** least-squares problem:

$$y = X\beta + \varepsilon, \qquad \hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y.$$

- The four estimators — **pooled, between, within, TWFE** — are this one formula applied to **four transformations of the data**: raw rows, unit means, deviations from unit means, and double-demeaned data.
- They differ only in **which variation in x** they exploit — *between* (across units, in the means $\bar{x}_i$) vs. *within* (over time inside a unit, $x_{it} - \bar{x}_i$) — hence in **what they must assume away**.
- The sampling variance is always the **sandwich**

$$\text{Var}(\hat{\beta}) = (X'X)^{-1}\, X'\Omega X\, (X'X)^{-1}, \qquad \Omega = \mathbb{E}[\varepsilon\varepsilon' \mid X],$$

and the whole inference question is: **what is Ω?** (we return to it).

Pooled OLS

Treat the panel as one big cross-section, leaving **both** unobserved effects in the error:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} = f_i + f_t + u_{it}.$$

- Removes **neither** f_i nor f_t from the error.
- **Consistent only if** $\text{Cov}(x_{it}, f_i) = 0$ **and** $\text{Cov}(x_{it}, f_t) = 0$ — usually too strong.
- ε_{it} shares f_i across all periods ⇒ **serially correlated** within i ⇒ cluster standard errors by i.

The bias, exactly

With a single regressor and exogenous idiosyncratic error ($\text{Cov}(x_{it}, u_{it}) = 0$), $\text{plim}\, \hat{\beta}_{\text{pooled}} = \beta + \dfrac{\text{Cov}(x_{it}, f_i + f_t)}{\text{Var}(x_{it})}$ — nonzero the moment x moves with *either* unobserved effect. The transformations below kill that covariance.

The between estimator

Run OLS of $\bar{y}_i$ on $\bar{x}_i$ — a *between* regression on the **unit means** (N data points, one per unit) with error $\bar{\varepsilon}_i$:

$$\bar{y}_i = \alpha + \beta\,\bar{x}_i + \bar{\varepsilon}_i, \qquad \hat{\beta}_B = \frac{\sum_i(\bar{x}_i - \bar{x})(\bar{y}_i - \bar{y})}{\sum_i(\bar{x}_i - \bar{x})^2}.$$

What sits in that error? Average the true model over t (f_i is constant, so it stays). In a **balanced panel** the time effects average to the same grand mean for every unit:

$$\bar{\varepsilon}_i = f_i + \bar{f} + \bar{u}_i, \qquad \bar{f} = \tfrac{1}{T}\textstyle\sum_t f_t.$$

- That $\bar{f}$ is the **same for all units** → a constant the intercept absorbs: **between sweeps out** f_t (its quiet advantage).
- f_i is **still in the error** → **biased if** $\mathrm{Cov}(\bar{x}_i, f_i) \neq 0$.
- Uses **only between** variation; a benchmark, rarely final. *(Unbalanced panel: the average differs across units and need not wash out.)*

What does each estimator remove from the error?

The error hides two unobserved effects, f_i and f_t. Each estimator differs by **which one it sweeps out**:

	removes f_i?	removes f_t?	left biased by
Pooled	no	no	f_i and f_t
Between	no	yes	f_i
Within	yes	no	f_t
TWFE	yes	yes	—

- **Between** averages over time $\rightarrow$ kills f_t, keeps f_i.
- **Within** demeans within units $\rightarrow$ kills f_i, keeps f_t (next).
- **TWFE** does **both —** the destination of this lecture.

*Between removes f_t only in a **balanced** panel; "Within keeps f_t" means in demeaned form, $f_t - \bar{f}$.*

Section 2

Fixed effects

The within transformation

Subtract each unit's time-average from the model:

$$y_{it} - \bar{y}_i = \beta\,(x_{it} - \bar{x}_i) + (f_t - \bar{f}) + (u_{it} - \bar{u}_i).$$

f_i cancels — but f_t does not

f_i is constant over t, so demeaning subtracts it from itself ($f_i - f_i = 0$): the individual effect **disappears**, and $\hat{\beta}_{\text{within}}$ is consistent even if $\text{Cov}(x_{it}, f_i) \neq 0$. But the time effect leaves $f_t - \bar{f} \neq 0$ in the equation.

- OLS on the demeaned data **is** the **within / fixed-effects** estimator,

$$\hat{\beta}_W = \frac{\sum_i \sum_t (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_i \sum_t (x_{it} - \bar{x}_i)^2},$$

  identified purely from **within-unit variation over time**.
- A common time shock can **still bias** it ⇒ we add **time effects** (TWFE) next.

Within and between: one variance, split in two

The two estimators are not rivals — they read **two halves of the same variance**. For a balanced panel, the total variation in x decomposes **exactly**:

$$\underbrace{\sum_{it}(x_{it}-\bar{x})^2}_{\text{total}} = \underbrace{\sum_{it}(x_{it}-\bar{x}_i)^2}_{\text{within (over time)}} + \underbrace{T\sum_{i}(\bar{x}_i-\bar{x})^2}_{\text{between (across units)}} .$$

- **Within** uses only the first piece, **between** only the second — same data, two different questions.
- Pooled OLS uses **both**: it is the variance-weighted average

$$\hat{\beta}_{\text{pooled}} = w\,\hat{\beta}_W + (1-w)\,\hat{\beta}_B, \qquad w = \frac{S_W}{S_W+S_B},$$

  with $S_W = \sum_{it}(x_{it}-\bar{x}_i)^2$ and $S_B = T\sum_i(\bar{x}_i-\bar{x})^2$ (the two pieces above — keep the T).
- **The price of within:** discarding the between piece ⇒ **larger standard errors**. Consistency is not free.

Balanced panel shown; unbalanced replaces T by T_i and weights accordingly.

Equivalence: least-squares dummy variables (LSDV)

Demeaning is identical to OLS with **one dummy per unit**: FE treats the unobserved f_i as **parameters to estimate** (one per unit), not as a random draw — each dummy's coefficient *is* f_i:

$$y_{it} = \beta x_{it} + f_i + u_{it}.$$

i	t	y_{it}	x_{it}	f_{USA}	f_{INDO}	f_{NIGER}
USA	2023	3.0	12	1	0	0
USA	2022	3.2	14	1	0	0
Indonesia	2023	1.0	22	0	1	0
Indonesia	2022	1.3	23	0	1	0
Niger	2022	0.1	28	0	0	1
Niger	2023	0.4	31	0	0	1

Same $\hat{\beta}$ as demeaning (Frisch–Waugh–Lovell). *Within* = looking **inside** each unit at its own evolution over time.

Frisch–Waugh–Lovell: within = partialling out f_i

Why are “demean” and “N dummies” the same answer? Collect the unit dummies in a matrix D and define the **residual-maker**

$$M_D = I - D(D'D)^{-1}D'.$$

- M_D is symmetric and idempotent[1] — a **projection**. Applied to any variable it **subtracts that variable’s unit means**: M_D *is* the within transformation.
- **Frisch–Waugh–Lovell:** regressing y on (X, D) gives the **same** $\hat{\beta}$ as regressing the residualized $M_D y$ on $M_D X$:

$$\hat{\beta}_W = (X' M_D X)^{-1} X' M_D y.$$

- So we never invert the huge $N \times N$ dummy block — we demean and run OLS on k columns. This is exactly what `feols`/`reghdfe` do internally.

[1] Idempotent: applying it twice equals applying it once, $M_D M_D = M_D$.

What within can — and cannot — do

Strict exogeneity (the key assumption)

$\mathbb{E}[u_{it} \mid x_{i1}, \ldots, x_{iT}, f_i] = 0$: the shock is uncorrelated with *past, present and future* regressors. (Rules out feedback, e.g. a lagged y on the right-hand side.)

- ✓ Controls for **all** time-invariant confounders, observed or not — they sit in f_i.
- × **Cannot estimate** the effect of any **time-invariant** regressor: if $x_{it} = x_i$ then $x_{it} - \bar{x}_i = 0$ and it drops out (gender, distance, “was colonized”…).
- × Still **vulnerable to time-varying** confounders — those are *not* in f_i.

Section 3

Two-way fixed effects

Two-way fixed effects (TWFE)

The within estimator left $f_t - \bar{f}$ in the equation — a **common shock** each period (crises, world prices, technology) can still bias it. Add a **time effect** f_t beside the unit effect:

$$y_{it} = \alpha + \beta x_{it} + \underbrace{f_i}_{\text{unit}} + \underbrace{f_t}_{\text{time}} + u_{it}.$$

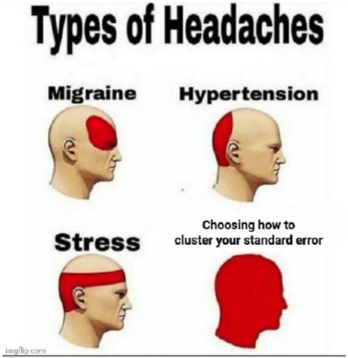
Estimate by **double-demeaning** ($\ddot{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}$, same for y) — FWL with both dummy sets:

$$\hat{\beta}_{TWFE} = \frac{\sum_{it} \ddot{x}_{it}\, \ddot{y}_{it}}{\sum_{it} \ddot{x}_{it}^2}.$$

- Absorbs time-invariant **unit** heterogeneity **and** unit-invariant **time** shocks.
- Stricter control of confounders $\rightarrow$ the **workhorse** of applied micro and the basis of difference-in-differences.
- Identified only if $\ddot{x}_{it}$ still **varies** (not collinear with the time dummies) — else the denominator collapses.

Section 4

Clustered standard errors



Source: Khoa Vu

The variance of FE estimators: where inference enters

We now have our point estimates — within and TWFE. The coefficient is settled; its **standard error** is not. By FWL any FE estimator is OLS on M_D-residualized data, so we just **replace X by $M_D X$** in the spine's sandwich:

$$\mathrm{Var}(\hat{\beta}_{\mathrm{FE}}) = (X' M_D X)^{-1} \big(X' M_D \, \Omega \, M_D X\big) (X' M_D X)^{-1},$$

where M_D sweeps out the fixed effects — unit dummies for within, **both** sets for TWFE.

- If $\Omega = \sigma^2 I$ this collapses to the textbook $\sigma^2 (X' M_D X)^{-1}$ — what default software prints.
- But within a unit the shocks u_{it} are **serially correlated**, so Ω is **not** scalar and that textbook formula **understates** the true variance.
- The honest fix is to model Ω as block-diagonal by unit — **clustered standard errors**, the rest of this section.

Why i.i.d. standard errors fail

Default software sets $\Omega = \sigma^2 I$ — **all errors independent**, equal variance. The previous slide showed why that breaks; here is the intuition:

- A unit's shocks **persist** across its years, so its own observations are **not** independent — the off-diagonal of Ω is nonzero.
- *Analogy:* counting 200 students who copied off 3 as 200 independent answers — you *think* you have 200 observations; you have a handful.
- Consequence: SEs **too small** $\Rightarrow$ t-stats inflated $\Rightarrow$ you **over-reject** (the stars lie).

Clustering: count blocks, not observations

Allow **any** correlation *within* a cluster (a unit across its years) but independence *across* clusters $\Rightarrow \Omega$ is **block-diagonal**, and

$$X'\Omega X = \sum_{g=1}^{G} X_g' \, \Omega_g \, X_g.$$

- The **cluster-robust** estimator replaces each block by the residual outer product (Liang–Zeger); consistent as the **number of clusters** $G \to \infty$ — *not* as the periods per unit grow.
- So **effective information = number of clusters** G**, not** NT**.**
- **Moulton factor:** if the regressor varies at the cluster level, the ratio of the true to the naïve variance is $1 + (\bar{n} - 1)\rho$, where $\bar{n}$ is the **cluster size** (observations per cluster) and ρ the **within-cluster correlation** of the errors. With $\bar{n} = 50$, $\rho = 0.1$: the right SE is $\sqrt{6} \approx 2.5\times$ the naïve one.

Where — and how many — to cluster

- **Cluster at the level of treatment assignment** (or wherever residuals stay correlated). With repeated observations per unit, cluster **by the unit**, not unit–year — the errors are serially correlated over time (Bertrand et al. 2004).
- **Coarser is safer** (weakly larger SE) — but you need **enough clusters**.
- **Too few clusters** $\Rightarrow$ the formula breaks: a **wild-cluster bootstrap** (simulate the null instead of trusting the t-table) restores the test — see the *Going further* appendix.
- Default: with grouped/repeated data, **cluster by the group — and report G.**

Section 5

Application

A long-standing question

Does a hot climate lower economic growth?

> *There are countries where the excess heat . . . makes men lazy and discouraged.*
> — *Montesquieu (1748)*

The modern version: high temperatures reduce productivity and growth. How do we test it **credibly**? (Dell et al. 2012)

Cross-section / between: what is missing?

A cross-section (or between) regression of growth on temperature,

$$\text{Growth}_i = \gamma \, \text{Temp}_i + \varepsilon_i,$$

compares **hot vs. cold countries —** which differ in many *fixed* ways (f_i):

- hot, poor countries were often **colonized**, with lasting institutional damage;
- the industrial revolutions happened in **temperate** climates (accident + slow diffusion).

These are **time-invariant confounders** → the cross-section / between estimate is biased.

Within: temperature shocks inside each country

Add a country fixed effect and use **year-to-year temperature variation within each country**:

$$\text{Growth}_{it} = \beta\,\text{Temp}_{it} + f_i + u_{it}.$$

- f_i absorbs colonization, institutions, geography — *every* fixed country trait.
- Identification: does a country grow **less in its own hotter years**? (data: many countries, 1950–2003.)
- Still a threat: **time-varying** confounders correlated with temperature (El Niño cycles, adaptation) — a country fixed effect cannot remove those.

Results: no effect on average

	Annual growth rate
Temperature	-0.325
	(0.285)

Country fixed effects; SE clustered by country.

- Negative but **statistically insignificant**: temperature fluctuations do not move growth *on average across all countries*.
- But an average can hide **heterogeneity**…

Heterogeneity: poor vs. rich

- Poor countries are concentrated in **agriculture** (up to ~80% of activity), poorly insured.
- Temperature should bite **harder** there than in rich, service-based, air-conditioned economies.
- Test it with an **interaction**: let the temperature effect differ for poor countries.
- We also add **year effects** f_t — global shocks (recessions, oil prices) common to all countries would otherwise contaminate the estimate.

Interaction: the effect in poor countries

With both country and **year** fixed effects (f_i, f_t):

$$g_{it} = f_i + f_t + \beta\, T_{it} + \alpha\, (T_{it} \times POOR_i) + u_{it}.$$

The marginal effect of temperature **depends on** $POOR_i$ — so you **add** the two coefficients for poor countries:

$$\frac{\partial g_{it}}{\partial T_{it}} = \beta + \alpha\, POOR_i = \begin{cases} \beta & \text{rich } (POOR = 0) \\ \beta + \alpha & \text{poor } (POOR = 1). \end{cases}$$

	Annual growth rate
Temperature	0.261 (0.312)
Temperature × Poor	−1.655*** (0.485)

So the poor-country effect is $\beta + \alpha = 0.261 - 1.655 = -\mathbf{1.394}$ pp — vs ≈ 0 in rich countries.

Country & year FE; clustered by country. Testing $\beta + \alpha$ needs its ***own*** *SE (a linear-combination test), not the stars on α.*

Section 6

Python

The toolkit and long format

Panel methods here use **`pyfixest`** — it *absorbs* fixed effects and clusters in one call (like Stata's `reghdfe` or R's `fixest`). Data must be in **long format**: one row per unit–period, with an `id` and a `time` column.

country	year	growth	temp
USA	2000	3.0	12
USA	2001	2.6	13
Niger	2000	0.5	28
Niger	2001	0.2	30

(Wide data — one column per year — must be **reshaped** to long first.)

Load, inspect, and check the panel

```python
import pandas as pd, pyfixest as pf

df = pd.read_stata("growth.dta")      # long: id, year, ...
df.groupby("country").size()          # periods per unit
df["country"].nunique()               # number of units N
```

- **Balanced** if every unit has the same count. **Singletons** (one period) carry no within variation and are dropped by the fixed effects.

Pooled → within → TWFE

```
cl = {"CRV1": "country"}     # cluster by unit

m_pool   = pf.feols("growth ~ temp", df, vcov=cl)
m_within = pf.feols("growth ~ temp | country", df, vcov=cl)
m_twfe   = pf.feols("growth ~ temp | country + year",
                    df, vcov=cl)
```

- `| country` absorbs f_i; `| country + year` absorbs f_i **and** f_t (TWFE).
- *(Between: regress the unit means — `df.groupby("country").mean()` then `feols` — rarely used.)*

Compare the estimators in one table

```
pf.etable([m_pool, m_within, m_twfe])    # side by side
```

- Watch $\hat{\beta}_{\text{temp}}$ **change** as each fixed effect enters — exactly the spine of this chapter (pooled keeps f_i, f_t; within drops f_i; TWFE drops both).

Heterogeneity and reading the output

```
m = pf.feols("growth ~ temp*poor | country + year",
             df, vcov=cl)
m.summary()      # coef, se, t, p; within-R2; #FE
```

- `temp*poor` expands to `temp + poor + temp:poor`; **`poor` is time-invariant**, so the country FE absorbs it — only `temp` and `temp:poor` are identified.
- The output also reports the **within-R^2** (fit from within variation) and how many fixed effects / observations were used.

Reporting, and a caveat

```
pf.etable([m_within, m_twfe], type="tex")   # LaTeX table
```

- Cluster-robust SEs need **many clusters**; with few, use a **wild-cluster bootstrap**.
- `pyfixest` ≈ Stata `reghdfe` ≈ R `fixest` — the same FE-absorption idea.

The same in R (fixest)

```r
library(fixest)
# within / TWFE / interaction, clustered by country
feols(growth ~ temp | country, df, cluster=~country)
feols(growth ~ temp | country + year, df, cluster=~country)
feols(growth ~ temp*poor | country + year, df,
      cluster=~country)
etable(m1, m2)          # to="tex" gives a LaTeX table
```

- Near-identical syntax: `y ~ x | FE1 + FE2`, `cluster = ~id`. Use whichever you prefer.

A caution: TWFE, DiD, and treatment timing

TWFE is the engine of difference-in-differences — but a warning for what comes next:

- With **staggered** treatment timing and **heterogeneous** effects, the TWFE DiD coefficient is a *contaminated* weighted average of effects — it can even be **negative** when every true effect is positive. *(A single-timing* 2×2 *DiD is unaffected.)*
- See Goodman-Bacon (2021) and de Chaisemartin–D'Haultfœuille (2020); newer estimators repair this.
- The **shift-share / IV** designs of the next course offer another route to credible causal effects.

Takeaways

- The error hides two unobserved effects — an individual f_i and a time f_t; panel methods **sweep them out**.
- **Pooled / between** use cross-sectional variation $\rightarrow$ **biased** under correlated heterogeneity.
- **Within (FE)** $\rightarrow$ **consistent** despite f_i, but cannot identify time-invariant regressors and needs **strict exogeneity**.
- **TWFE** also nets out common time shocks — the workhorse, and the bridge to DiD.
- FE/TWFE removes confounders **en bloc**: it nets out f_i and f_t without telling us *which* underlying variables they stood for.
- Always **cluster** standard errors (usually by unit) — and check you have enough clusters.

References

For a fuller textbook treatment of every method in this chapter, see Wooldridge (2025).

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *The Quarterly Journal of Economics* 119 (1): 249–75. https://doi.org/10.1162/003355304772839588.

Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken. 2012. "Temperature Shocks and Economic Growth: Evidence from the Last Half Century." *American Economic Journal: Macroeconomics* 4 (3): 66–95. https://doi.org/10.1257/mac.4.3.66.

Wooldridge, Jeffrey M. 2025. *Introductory Econometrics: A Modern Approach*. 8th ed. Cengage Learning.

Section 7

Appendix

The covariance Ω: what we assume

Everything is the **sandwich** $\text{Var}(\hat{\beta}) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$, with $\Omega = \mathbb{E}[\varepsilon\varepsilon' \mid X]$ the $NT \times NT$ error covariance. Only the **meat** $X'\Omega X$ changes with our assumption:

Three choices for Ω — same $\hat{\beta}$, different SE

i.i.d.: $\Omega = \sigma^2 I$ (classical). **Heteroskedastic:** $\Omega = \text{diag}(\sigma_i^2)$ (White/HC). **Clustered:** Ω block-diagonal (free within groups).

The point estimate never moves; the standard error is entirely a statement about Ω.

The cluster-robust estimator (Liang–Zeger, 1986)

With G clusters — errors free *within*, independent *across*:

$$X'\Omega X = \sum_g X_g'\Omega_g X_g \approx \sum_g X_g' \hat{\varepsilon}_g \hat{\varepsilon}_g' X_g,$$

$$\widehat{\mathrm{Var}}_{\mathrm{CR}}(\hat{\beta}) = \tfrac{G}{G-1}\,(X'X)^{-1}\Big(\textstyle\sum_g X_g'\hat{\varepsilon}_g\hat{\varepsilon}_g' X_g\Big)(X'X)^{-1}.$$

- Each cluster's residual outer product is a *poor* estimate of Ω_g — but the **sum over independent clusters** is consistent. $\frac{G}{G-1}$ is a small-sample fix.
- Consistent as $G \to \infty$ (**many clusters**), *not* as the cluster size grows.

Moulton (1990): the canonical lesson

A cluster-level regressor with equicorrelated errors (intraclass correlation ρ, cluster size $\bar{n}$) gives the **Moulton factor**:

$$\frac{\mathrm{Var}(\hat{\beta})_{\text{true}}}{\mathrm{Var}(\hat{\beta})_{\text{naive}}} = 1 + (\bar{n} - 1)\rho.$$

- $\bar{n} = 50, \rho = 0.1 \Rightarrow \sqrt{6} \approx 2.5\times; \bar{n} = 1000, \rho = 0.05 \Rightarrow \approx 7\times.$
- A country-level tax rate with 30{,}000 firms: information lives at the **country**, not the firm. You cannot escape clustering by disaggregating.

Bertrand–Duflo–Mullainathan (2004): the experiment

A famous stress-test of clustered inference. The question: do the **standard standard errors** used in policy evaluations actually have the right size?

- **Data.** The *Current Population Survey* (CPS) — the large monthly U.S. household survey — gives many years of individual wages across the 50 states.
- **Placebo law.** They invent a **fake** state policy: pick a random state and a random year, and pretend a law switched on there. It changes nothing, so the true effect is **zero by construction** ($\beta = 0$).
- **The test.** Estimate the "effect" of this fake law on wages and ask how often the t-test rejects $\beta = 0$ at the 5% level. With honest SEs it should reject $\approx 5\%$ of the time. Repeat over hundreds of random placebos.

Bertrand–Duflo–Mullainathan (2004): the trap, and the fix

The result

Classical (i.i.d.) SEs rejected the true null in $\approx 45\%$ of placebos — nine times the nominal 5%. Clustering by **state** brought it back to 6–8%.

- **Why so bad?** Wages are **serially correlated** within a state (this year looks like last year), *and* the fake law, once on, **stays on** for years (treatment is persistent). The many state–year rows are far from independent — effectively there is only **one experiment per state**, not one per state–year.
- **The fix.** Cluster at the **state** level (the unit of treatment): the SEs then count ≈ 50 blocks, not thousands of rows, and the test size is restored.
- **Lesson.** Cluster at the level of treatment assignment — not the finer unit–year.

Abadie–Athey–Imbens–Wooldridge (2023): when to cluster, really

Reframes clustering as a question about **design**, not residuals. Cluster if **either**:

1. **Sampling design —** you sampled clusters, then units within them (schools, then pupils).
2. **Assignment design —** treatment is assigned at the cluster level.

- If **neither** holds (a census; unit-specific treatment), clustering may be unnecessary and can even *cost* precision.
- "Are errors correlated?" is the wrong question; "did the data or the treatment arrive in clusters?" is the right one.

Where to cluster: the practical rule

Rule

Cluster at the **coarsest** level at which (i) treatment is assigned / sampling was done, or (ii) residuals stay correlated.

- Nested clusterings $C_1 \subset C_2$: $\widehat{\mathrm{Var}}_{C_2} \geq \widehat{\mathrm{Var}}_{C_1}$ — **coarser is (weakly) larger**, hence safer for test size.
- But coarser shrinks G; below $\sim$ 15–20 clusters the formula degrades.
- **Never cluster finer than the level of treatment assignment.**

Multiway clustering (Cameron–Gelbach–Miller, 2011)

Correlation along **two** non-nested dimensions (e.g. country *and* year):

$$\widehat{\text{Var}}_{\text{2-way}} = \widehat{\text{Var}}_{\text{country}} + \widehat{\text{Var}}_{\text{year}} - \widehat{\text{Var}}_{\text{country} \cap \text{year}}.$$

- Each piece is a valid one-way CRV; subtracting the intersection avoids double-counting.
- Use only with a clear reason for two dimensions; **unstable** when one dimension has few clusters (e.g. few years).

Few clusters: two distinct problems

CRV1 is consistent only as $G \to \infty$. When G is small it misbehaves in **two** ways:

1. **Reference distribution.** Compare t to t_{G-1}, not the normal (e.g. $t_{21,\,0.975} = 2.08$ vs 1.96) — software does this automatically.
2. **Noisy variance.** $\sum_g X_g' \hat{\varepsilon}_g \hat{\varepsilon}_g' X_g$ is itself a noisy estimate and **under-covers**, worse with unbalanced clusters.

Each needs a *different* remedy.

Few clusters: the remedies

Rather than trust the t-table when G is small, **simulate** the test's own distribution and read the p-value off it.

- **Wild-cluster bootstrap** (Cameron–Gelbach–Miller 2008): re-estimate the model thousands of times, each time flipping the sign of every cluster's residuals at random (± 1). This builds the distribution of the t-statistic *under the null*, using only G blocks. The historic default — but it **fails with few *treated* clusters** (can trap $p \approx 1$). Use with care.
- **CR2 + Satterthwaite df** (Imbens–Kolesár 2016; Pustejovsky–Tipton 2018): bias-corrected sandwich + effective df. Good default to $G \approx 10$ when leverage is balanced.
- **CR3 cluster jackknife** (MacKinnon–Nielsen–Webb 2023): robust when a few clusters carry most of the leverage.
- **Randomization inference** (Young 2019): permute treatment across clusters; distribution-free.

Summary, and further reading

Decision tree

Assigned/sampled in clusters, or residuals correlated? $\rightarrow$ cluster there. Coarser $=$ safer, but keep enough clusters. Few clusters $\rightarrow$ CR2 / CR3 or randomization inference, **not** a naive wild bootstrap when treatment is concentrated.

Moulton (1990); Liang–Zeger (1986); Bertrand–Duflo–Mullainathan (2004); Cameron–Gelbach–Miller (2008, 2011); Cameron–Miller (2015); Imbens–Kolesár (2016); Pustejovsky–Tipton (2018); Abadie–Athey–Imbens–Wooldridge (2023); MacKinnon–Nielsen–Webb (2023); Young (2019).